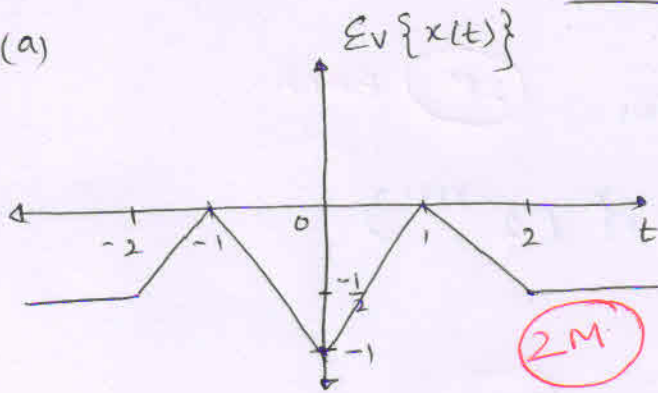
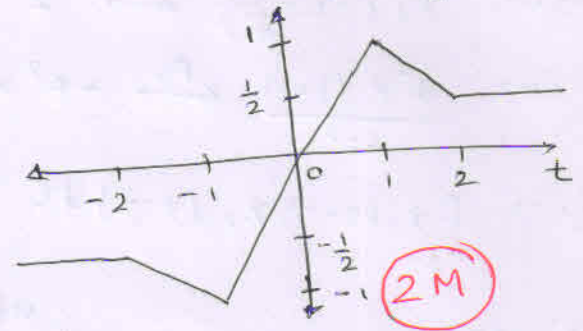


Mid-Term Solutions

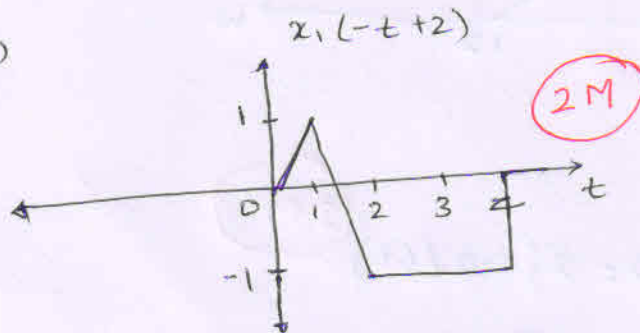
1) (a)



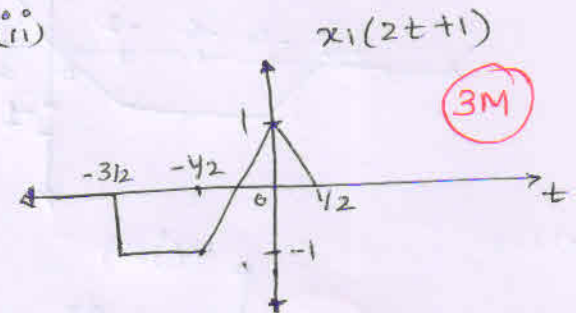
$od\{x(t)\}$



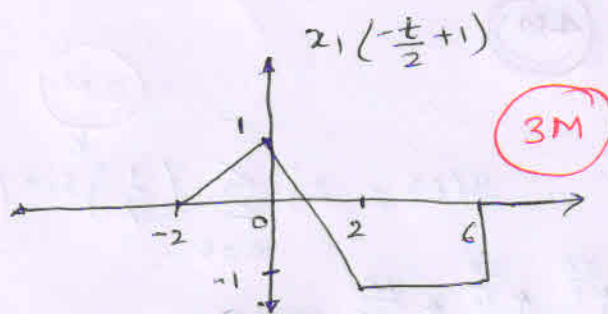
(b) (i)



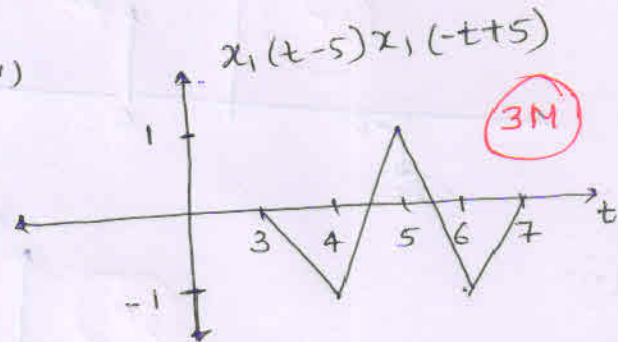
(ii)



(iii)



(iv)



2) (a)

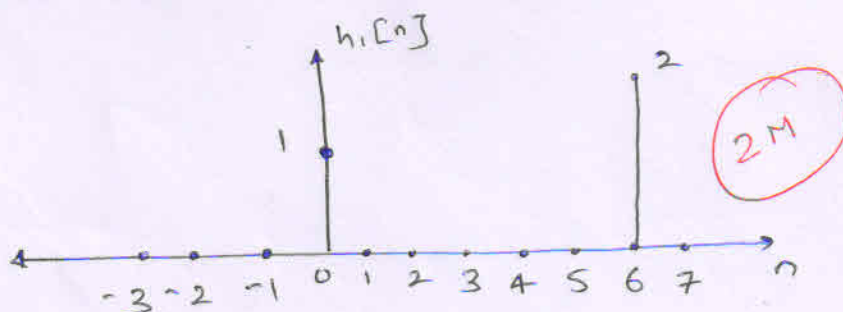
(i) $s(t) = (1 - e^{-(t-2)})u(t-2)$ 3M

(ii) $y(t) = \begin{cases} 0 & t < 0 \\ 1 - e^{-t} & 0 < t < 3 \\ e^{-(t-3)} - e^{-t} & t > 3 \end{cases}$ 6M

(b)

(i) $y_1[n] = 2x[n] - 2x[n-6]$ 4M

(ii)



Q3) (a) a_2 and $a_{-2} = 0$, a_1 and $a_{-1} \neq 0$ — (3M)

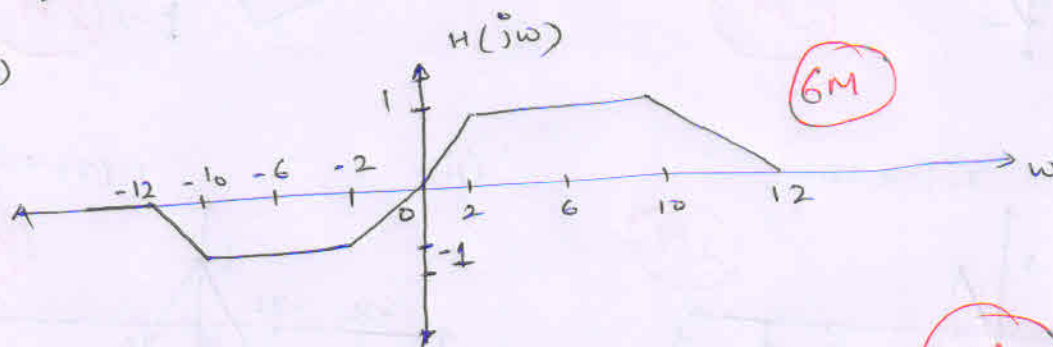
$x(t) = \frac{1}{5} \cos\left(\frac{\pi}{5}t\right)$ — (3M)

(b) (i) $x_1\left(-t + \frac{1}{4}\right) \xleftrightarrow{FS} 2^{-|k|} e^{-j\frac{k\pi}{4}}$ (2M) Each.

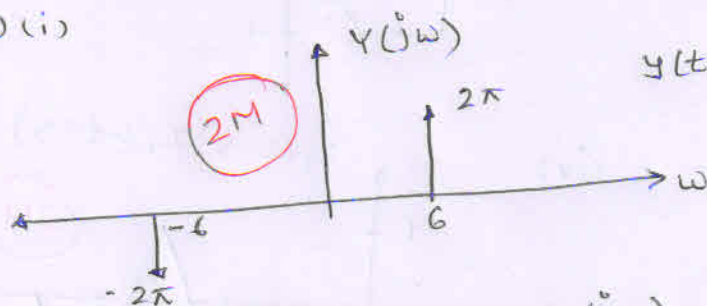
(ii) $\frac{d^2 x_1(t+1)}{dt^2} \xleftrightarrow{FS} -k^2 \pi^2 (-1)^k 2^{-|k|}$

(iii) $\int_2 x_1(t-1) x_2(t-1) dt \xleftrightarrow{FS} 2(-1)^k (2^{-2|k|})$

Q4) (a)

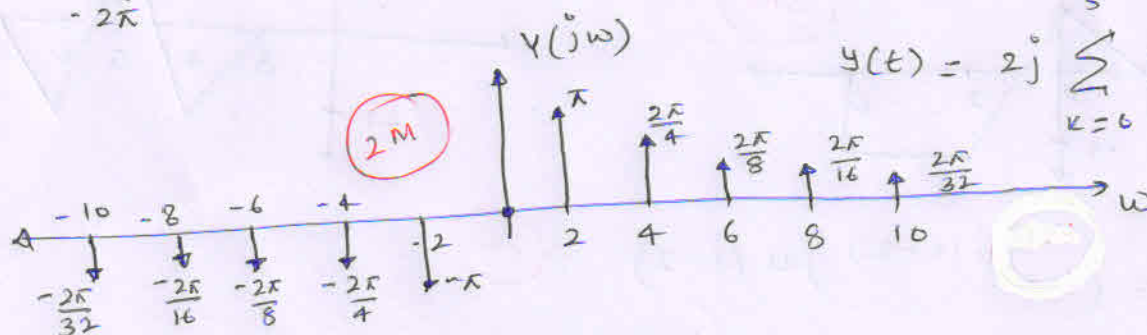


(b) (i)



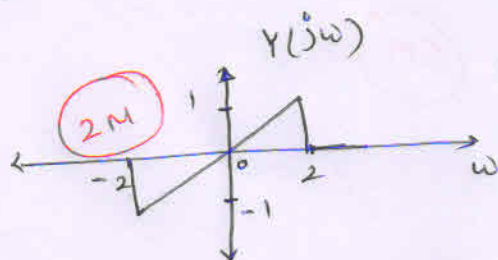
$y(t) = 2j \sin(6t)$ (2M)

(ii)



$y(t) = 2j \sum_{k=6}^5 \left(\frac{1}{2}\right)^k \sin(2kt)$ (2M)

(iii)



$y(t) = \frac{j}{2} \left(\frac{\sin(2t) - 2t \cos(2t)}{\pi t^2} \right)$ (2M)